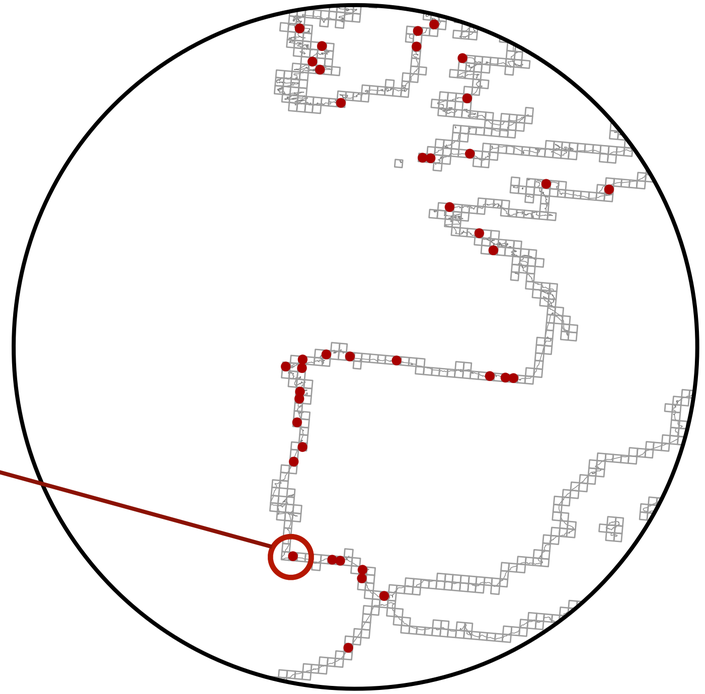
Session 3: The Architecture of Species Distribution Models

Modeling the distribution of biodiversity under global climate change

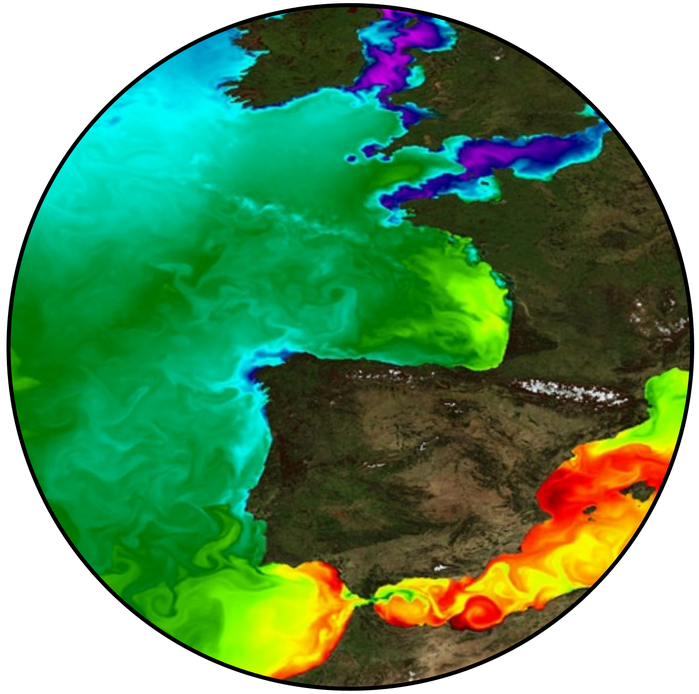
Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

Data Preparation

Obs.	SST	Salinity	Oxygen
1	**21.0**	**35.4**	**210.1**
1	19.4	29.1	224.9
1	21.5	31.1	199.8
0	21.3	33.4	201.1
0	14.8	34.1	184.8
(…)	(…)	(…)	(…)



Records of species distribution



Environmental data

Linking Biology to Environment: from Maps to Matrices

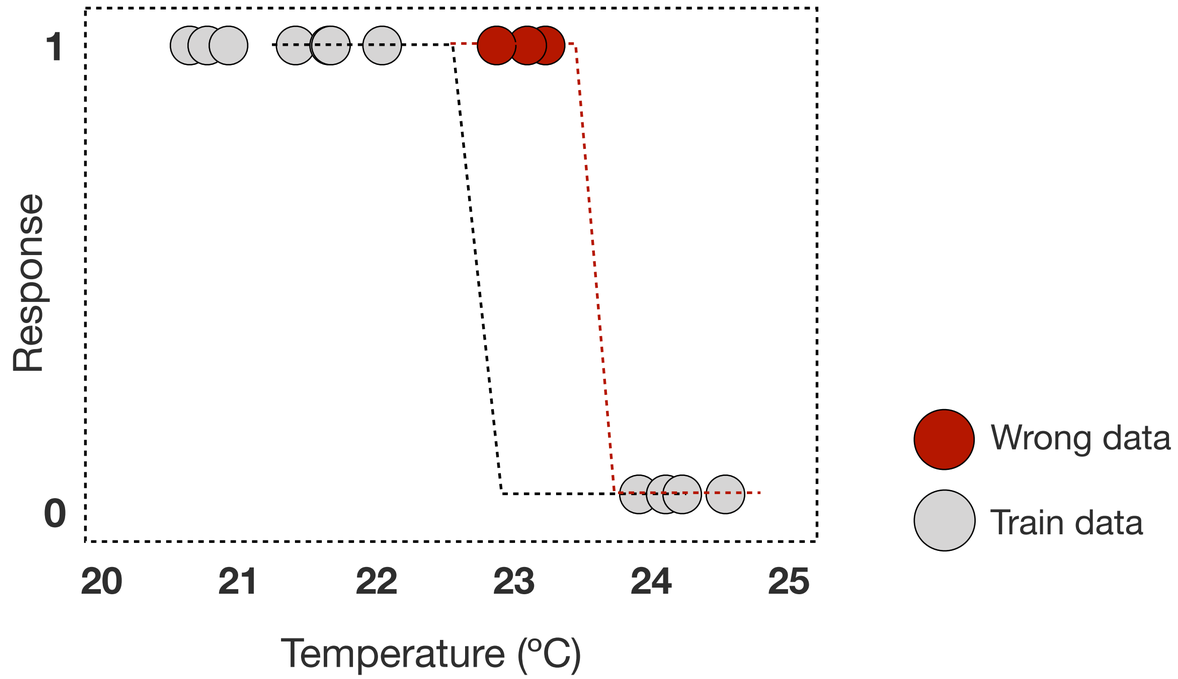
Algorithms cannot read maps; they require a data matrices. Achieved by overlaying vector data (**species occurrences**) onto raster grids (**environmental layers**) - extracting the exact pixel values at those coordinates.

"Garbage In, Garbage Out"

Model outputs are only as good as the input data

The quality of several sources (e.g., GBIF) has been questioned.

There are **inaccuracies** and **bias** in occurrence records.

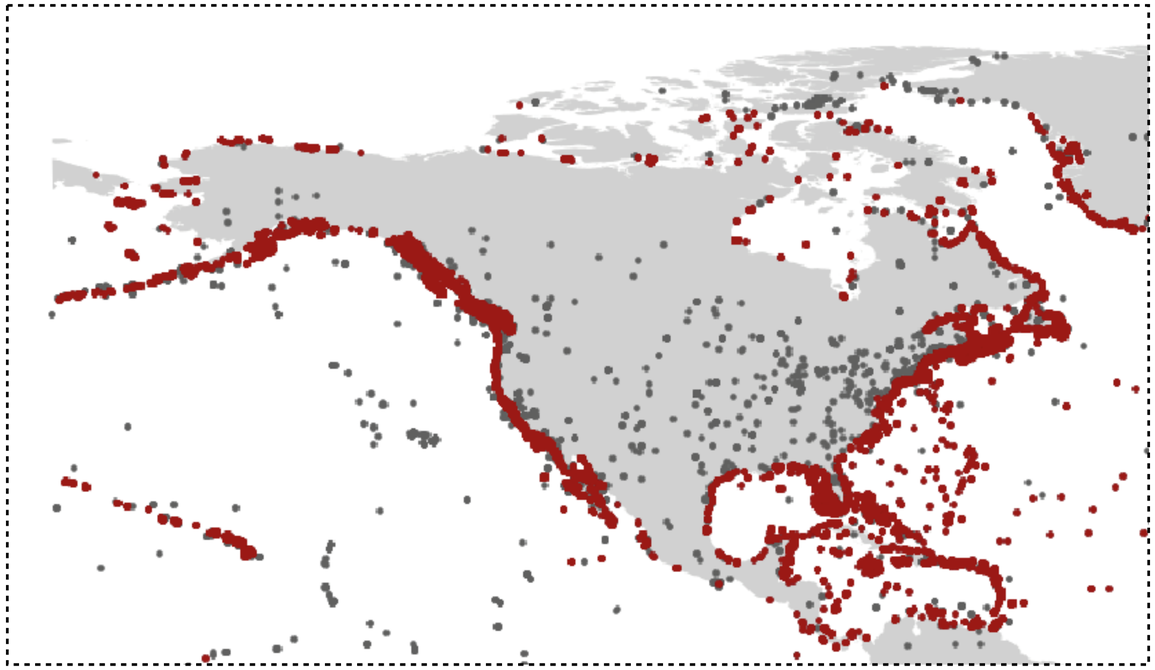


Inaccurate records

Records outside the distribution of the species

Problem: the models may capture **a broader niche**, i.e., the span of conditions which the specie is exposed to.

Manual **curation is needed** prior to modeling - the molders needs to **know the distribution of the species** to properly curate the records - *the geographic and vertical distribution (e.g., marine realm).*



Inaccurate records

Marine records over land (and vice versa)

Problem: Wrong records that **do not provide relevant information** to the model.

Can also arise from coordinate imprecision, flipped Lon-Lat or wrong source information (the museum location instead of records). The modeler needs to discard unreliable coordinate data.

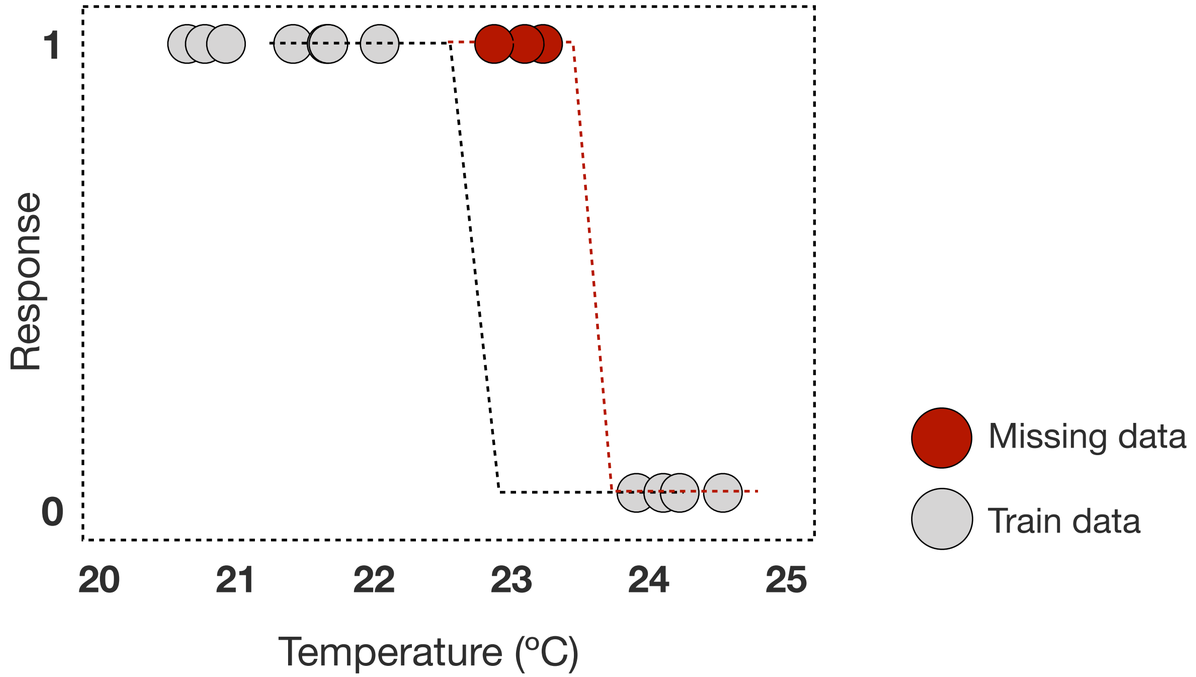
How many presence occurrence records?

Sample size can be positively correlated to the performance of models.

Experiments point to **at least 50 records to estimate response functions, with at least 10 per variable.**

The **number of observations is less important than having well-distributed observations** - cover the environmental space where the species is distributed. Thus, compiling records should be guided by knowledge on species distributions.

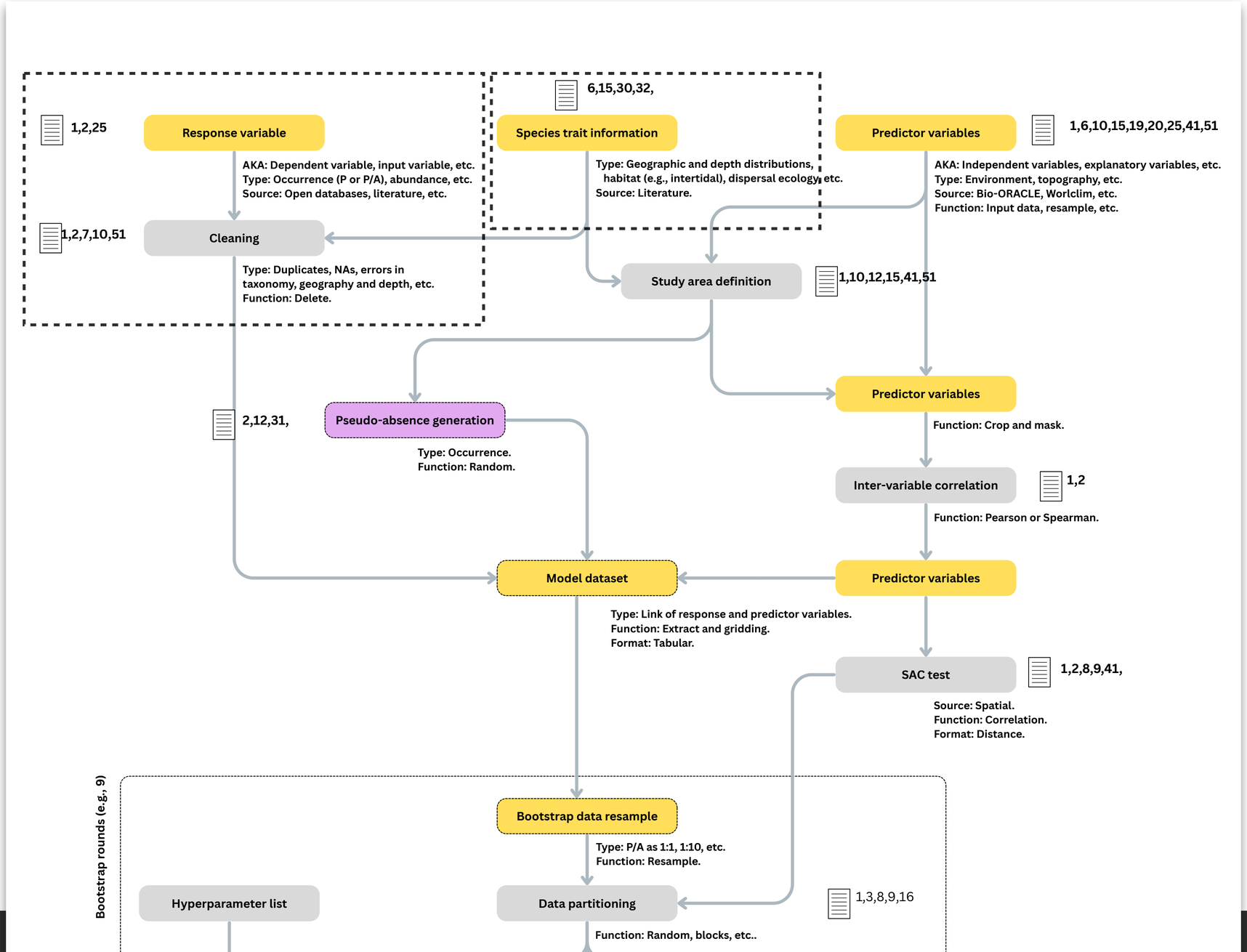
Fei & Yu, 2015

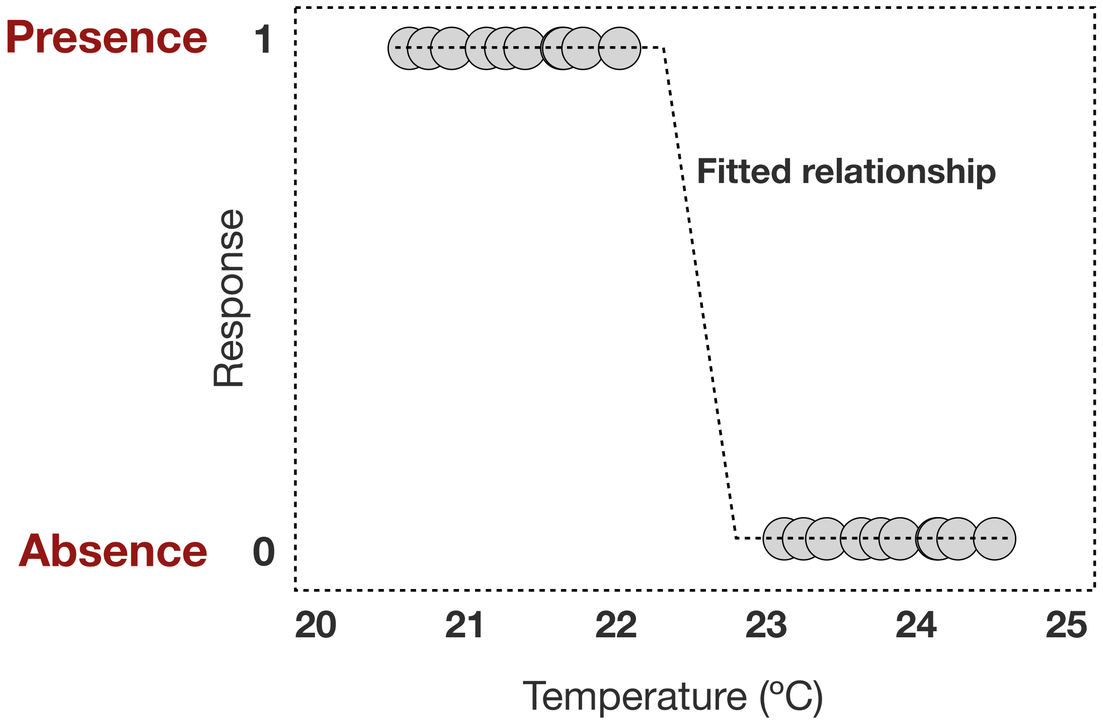


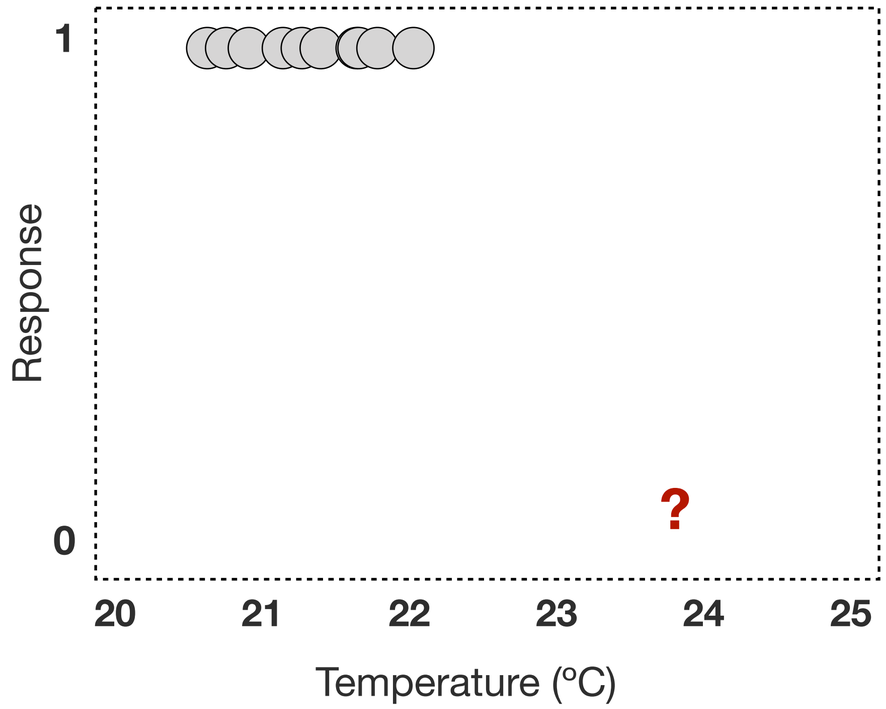
Niche truncation

If presence records are missing, the models may not capture the full width of the niche, i.e., the span of conditions which the specie is exposed to.

The model fits the portion of the niche represented by the observed records.


1,2,25
Response variable
AKA: Dependent variable, input variable, etc.
Type: Occurrence (P or P/A), abundance, etc.
Source: Open databases, literature, etc.
6,15,30,32,
Species trait information
Type: Geographic and depth distributions, habitat (e.g., intertidal), dispersal ecology, etc.
Source: Literature.
Predictor variables
1,6,10,15,19,20,25,41,51
AKA: Independent variables, explanatory variables, etc.
Type: Environment, topography, etc.
Source: Bio-ORACLE, Worlclim, etc.
Function: Input data, resample, etc.
1,2,7,10,51
Cleaning
Type: Duplicates, NAs, errors in taxonomy, geography and depth, etc.
Function: Delete.
Study area definition
1,10,12,15,41,51
Predictor variables
Function: Crop and mask.
2,12,31,
Pseudo-absence generation
Type: Occurrence.
Function: Random.
Inter-variable correlation
1,2
Function: Pearson or Spearman.
Predictor variables
Model dataset
Type: Link of response and predictor variables.
Function: Extract and gridding.
Format: Tabular.
SAC test
1,2,8,9,41,
Source: Spatial.
Function: Correlation.
Format: Distance.
Bootstrap rounds (e.g., 9)
Bootstrap data resample
Type: P/A as 1:1, 1:10, etc.
Function: Resample.
Hyperparameter list
Data partitioning
1,3,8,9,16
Function: Random, blocks, etc..





Handling Presence-Only Data

Correlative models infer **the relationship between species presence and absence records and the environmen**t, but **absences are often unavailable or unknown**, leading to **presence-only datasets** (common for marine species).

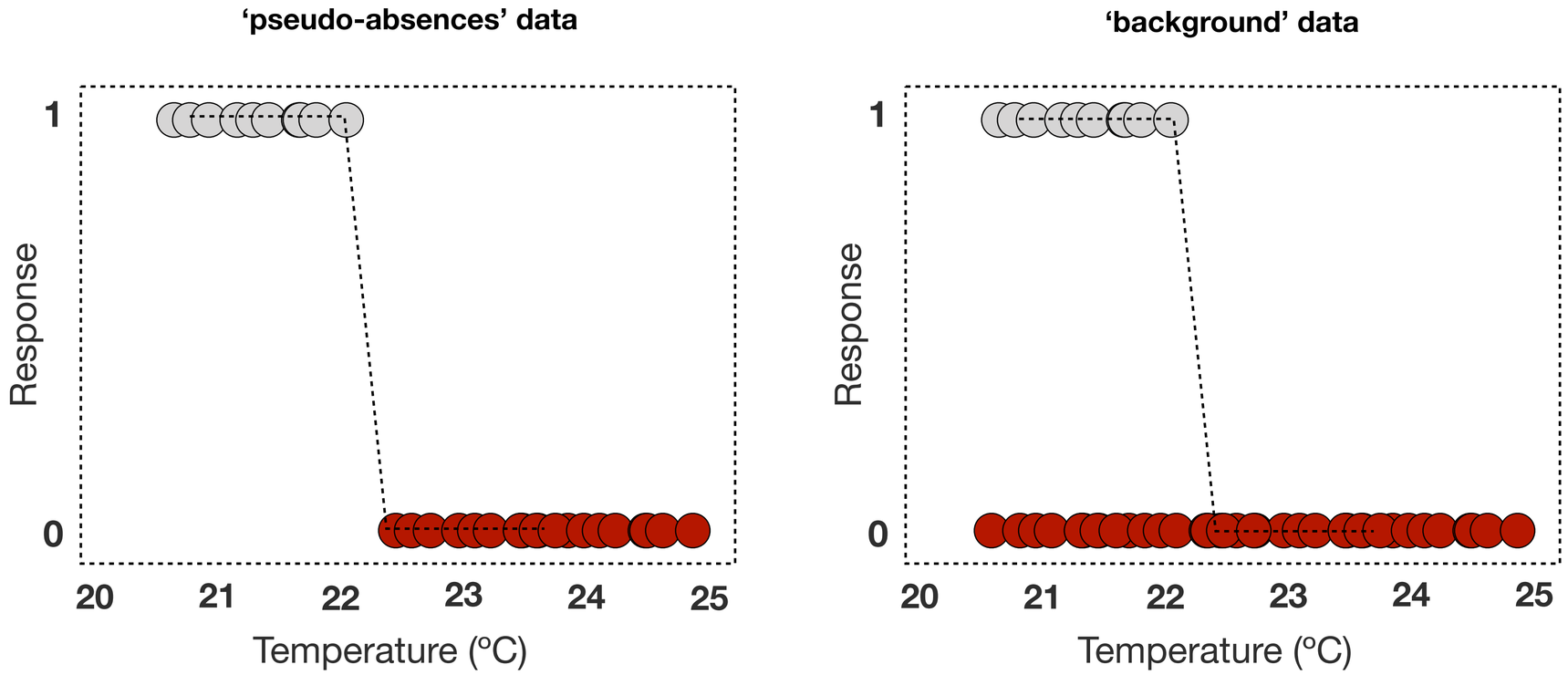
Detectability bias

The absence of a recorded does not mean the absence of the species.

A species **might be present but unobserved** due to cryptic behavior, daily migrations, or simply a lack of survey effort.

Some species more easily detected. The **more mobile the species, the greater is the detection problem**.

Problem: Treating un-surveyed areas as "true absences" severely penalizes the process of capturing the niche, *making presence-only techniques necessary*.



Strategies for presence-only modeling

Pseudo-absences: Artificially generated in sites of the study area where presences have not been recorded. Regression algorithms can be implemented (e.g., GLM and BRT).

Background data: Artificially generated from the entire study area. Focuses on comparing the conditions where the species occurs vs. the entire study area (e.g., MaxEnt).

How many absence records?

Pseudo-absences:

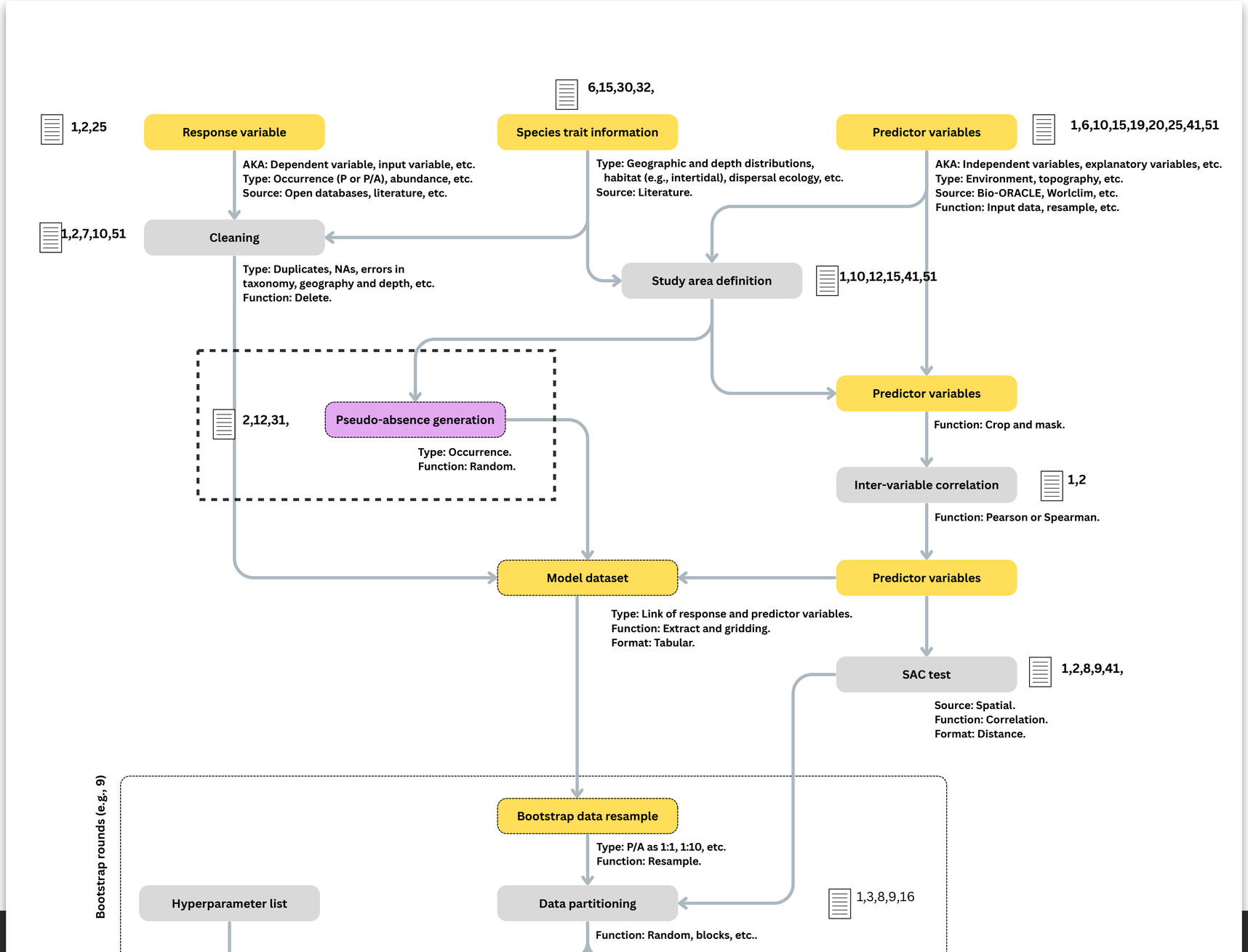
Linear models: a large number (typicality 10,000).

Advances machine learning models (e.g., BTR and XGBoost): the same number as presence records for balanced datasets (when less than 1000 P, at least 100 PA).

Background data:

Maximum entropy (MaxEnt): a large number (typicality 10,000).

Barbet-Massin et al., 2012


6,15,30,32,
1,2,25
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Species trait information
Predictor variables
1,6,10,15,19,20,25,41,51
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Function: Resample.
Hyperparameter list
Data partitioning
1,3,8,9,16
Function: Random, blocks, etc..

Environmental variables: selection

Ecological relevance is key:

Choose variables **known or hypothesized to directly influence the species' survival, growth, or reproduction**. Consider physiology and life history (literature).

Using large datasets is an approach with stronger criticism.

Important marine variables:

Physical: temperature, salinity, bathymetry, currents velocity, sea ice conditions.

Chemical: oxygen, nutrients (e.g., nitrate and phosphate).

Biological: chlorophyll-a (proxy for productivity).

Zurell et al., 2020

		Present	SSP1-1.9	SSP1-2.9	SSP2-4.5	SSP3-7.0	SSP4-6.0	SSP5-8.5
		2 (2000-2020)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)
1	**Temperature**	■	■	■	■	■	■	■
2	**Salinity**	■	■	■	■	■	■	■
3	**Sea Ice Cover**	■	■	■	■	■	■	■
4	**Sea Ice Thickness**	■	■	■	■	■	■	■
5	**Sea Water Velocity**	■	■	■	■	■	■	■
6	**Mixed Layer Depth**	■	■	■	■	■	■	■
7	**Diffuse Attenuation Coefficient**	■						
8	**PAR**	■						
9	**PAR at Bottom**	■						
10	**Oxygen**	■	■	■	■	■	■	■
11	**pH**	■	■	■	■	■	■	■
12	**Iron**	■	■	■	■	■	■	■
13	**Phosphate**	■	■	■	■	■	■	■
14	**Nitrate**	■	■	■	■	■	■	■
15	**Silicate**	■	■	■	■	■	■	■
16	**Primary Production**	■	■	■	■	■	■	■
17	**Phytoplankton**	■	■	■	■	■	■	■
18	**Chlorophyll**	■	■	■	■	■	■	■

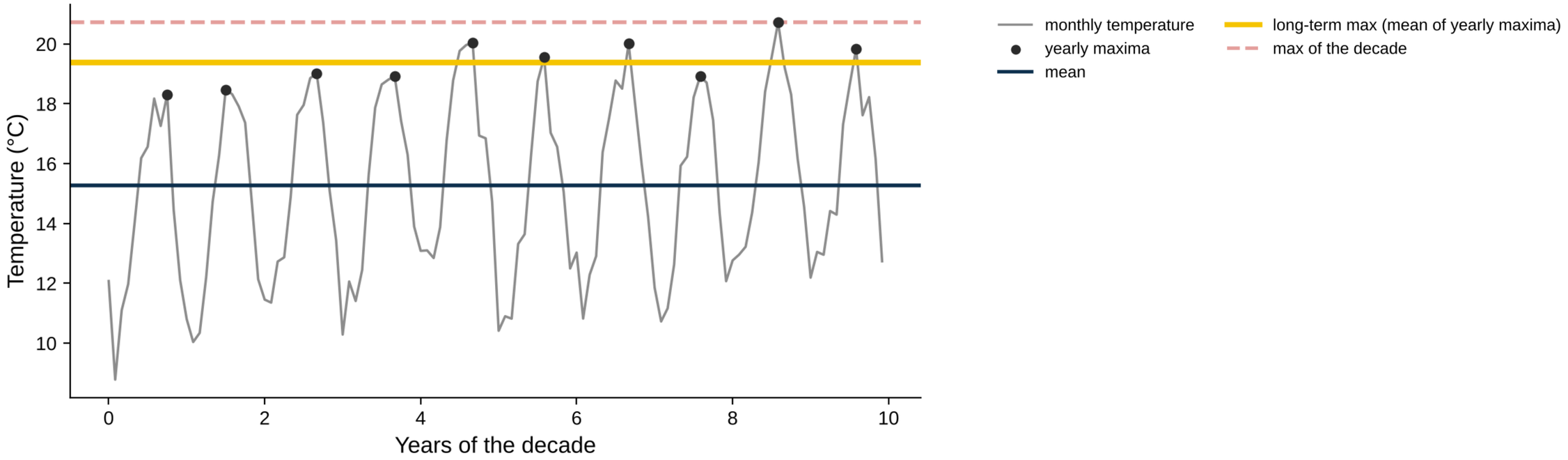
Bio-ORACLE: **18 essential variables** for the surface and along the seafloor, **from 2000 to 2100**, **per decade**, at ~5km **resolution**, **covering the span of SSPs**, by **ensembling 10 ESMs of CMIP6**.

Six statistics per variable

Per decade and variable: the mean; the maximum and minimum of the decade; the long-term maximum and minimum (the average of the yearly extremes); and the range.

Rule: limits on survival call for extremes (long-term max or min).

A kelp that dies above a temperature threshold: mean less important than long-term max.



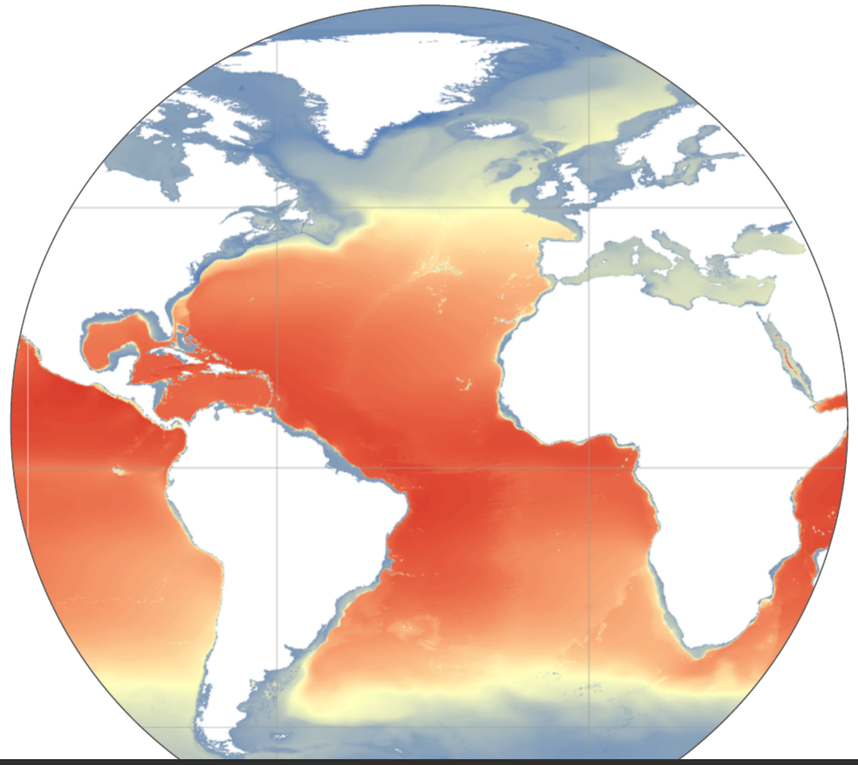
Assis et al., 2024

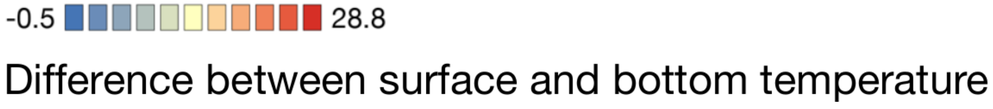
Surface or sea floor

The 3D ocean: marine species live across depth gradients, unlike most terrestrial species.

Surface layers: for pelagic and very shallow species.

Benthic layers: for species on the bottom, e.g., corals or demersal fish.





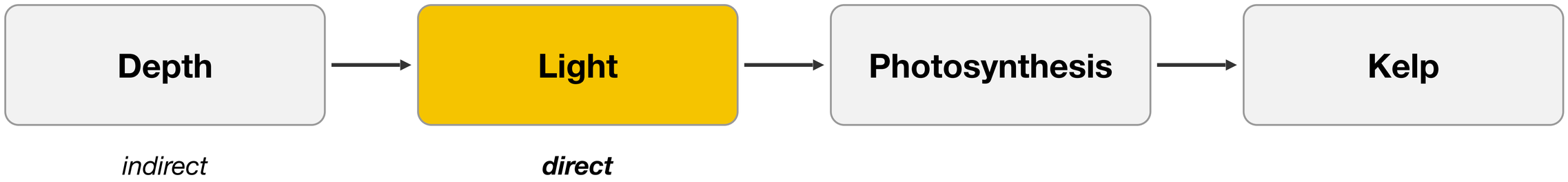
Assis et al., 2024

Ecological relevance first

Direct (proximal) predictors: act on survival, growth or reproduction, e.g., maximum temperature against a lethal limit, light for photosynthesis, oxygen for respiration.

Indirect predictors: correlate with the causes, e.g., latitude, or depth as a stand-in for light.

Why it matters: direct predictors keep their meaning when the model is projected to new places or times; indirect ones may not.



2 min. PITCH

Papers: P49, P61

10 minutes

Read the abstract and the introduction and interpret the main figures (P49 Figs 1,2; P61 2,3).

Question. What problem did the authors tackle? (one sentence)

Approach. What did they do, and with which data and model? (one sentence)

Key finding. The one result to remember, ideally with the paper's main figure.

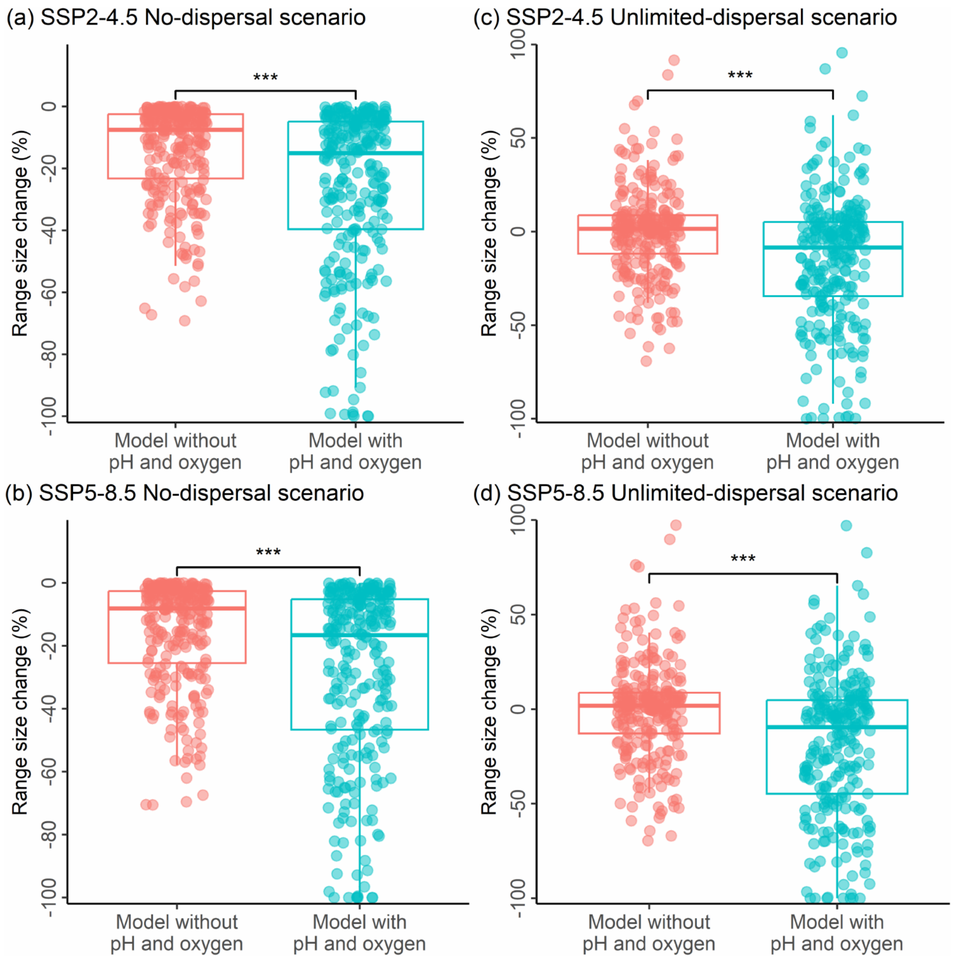
Why it matters today?

Beyond temperature

Most projections overlook pH and dissolved oxygen.

Cold-water corals, 268 species: models with pH and oxygen projected 11.5 to 21.4% higher impacts than models without them.

By 2100, high emissions, no dispersal: a mean range contraction of 48.2% with pH and oxygen, against 26.8% without.



Liu et al., 2026

Environmental variables: selection

Marine mammals (e.g., whales, dolphins, seals):

Often driven by **temperature tolerances, prey distribution, and physical barriers** (e.g., sea ice).

Potential variables:

Ocean temperature: physiological tolerance limits.

Primary productivity / Chlorophyll: proxy for food web base supporting prey.

Bathymetry: depth limits for diving / foraging, habitat association.

Sea ice cover / sea ice thickness: Critical habitat for polar species (e.g., seals, polar bears).

Environmental variables: selection

Fish (e.g., Demersal Cod, Pelagic Tuna):

Temperature and oxygen are critical physiological drivers, plus factors related to food.

Potential variables:

Ocean temperature: physiological tolerance limits.

Salinity: physiological tolerance limits.

Dissolved molecular oxygen: physiological tolerance limit, especially in deeper or oxygen minimum zones.

Bathymetry: strong determinant of habitat zones (e.g., demersal species).

Primary productivity / Chlorophyll: proxy for food web base supporting prey.

Environmental variables: selection

Primary producers like macroalgae and seagrasses:

Requirements for light, nutrients, suitable substrate, temperature limits, water motion.

Potential variables:

Ocean temperature: physiological tolerance limits.

Salinity: physiological tolerance limits.

Photosynthetically Available Radiation (PAR): Essential for photosynthesis.

Diffuse attenuation: Measures light penetration depth, restricts photosynthesis.

Nitrate, Phosphate, …: Key nutrients for growth.

Environmental variables: selection

Corals (e.g., warm-water Reef Builders, cold-water Corals):

Temperature sensitivity, light (for symbiotic warm-water corals), substrate, water flow.

Potential variables:

Ocean temperature: physiological tolerance limits.

Salinity: physiological tolerance limits.

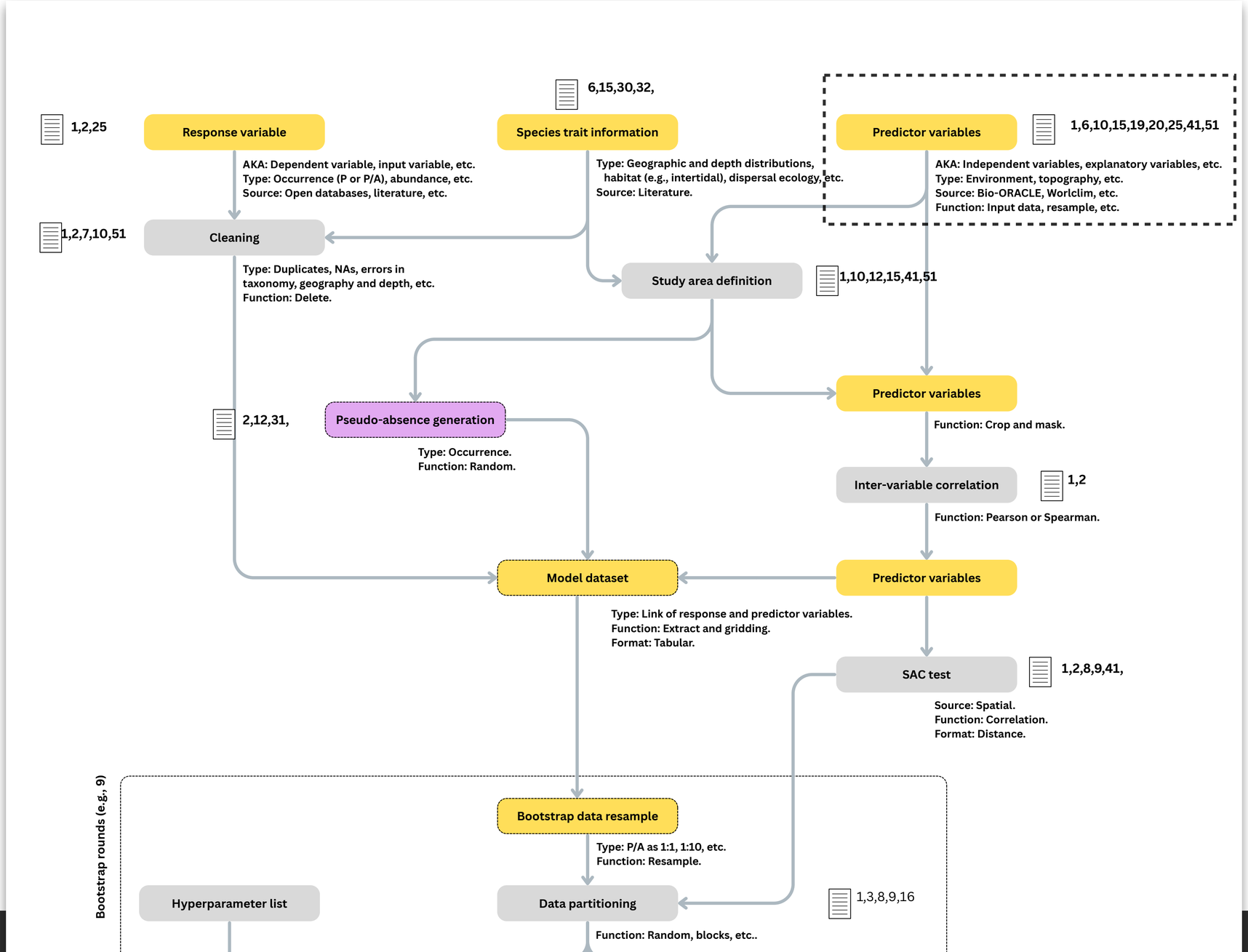
pH: affects calcification (building skeletons).

Dissolved molecular oxygen: physiological tolerance limits.

Bathymetry: defines depth range.

(For symbiotic zooxanthellate) PAR, Diffuse attenuation: Light for photosynthesis.

(For azooxanthellate) Primary productivity / Chlorophyll: Proxy for food supply.


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Defining the study area

Calibration vs. projection area:

Calibration area: where the model is trained (i.e., where the data is distributed).

Projection area: where the model is applied to make predictions (can be broader, e.g., globally, future climates, new regions for invasion risk).

The **choice of calibration area significantly influences the model**, and therefore performance and transferability.

Defining the study area

Major question while defining the calibration area: Which areas has the species been able to access over time? (the species' accessible area, the 'M' in BAM).

The area must comprise the accessible areas to capture the range of conditions the species has experienced (suitable and unsuitable).

Rule of thumb: choose an area relevant to the species' dispersal ecology, large enough to capture the environmental gradients.

e.g.,

The larvae dispersal potential of a coral species is 10 days; in the ocean this can be translated onto 200 km dispersal potential - the study area is defined as the area around any observed record using a buffer of 200km.

www.nature.com/scientificdata

scientific data

Check for updates

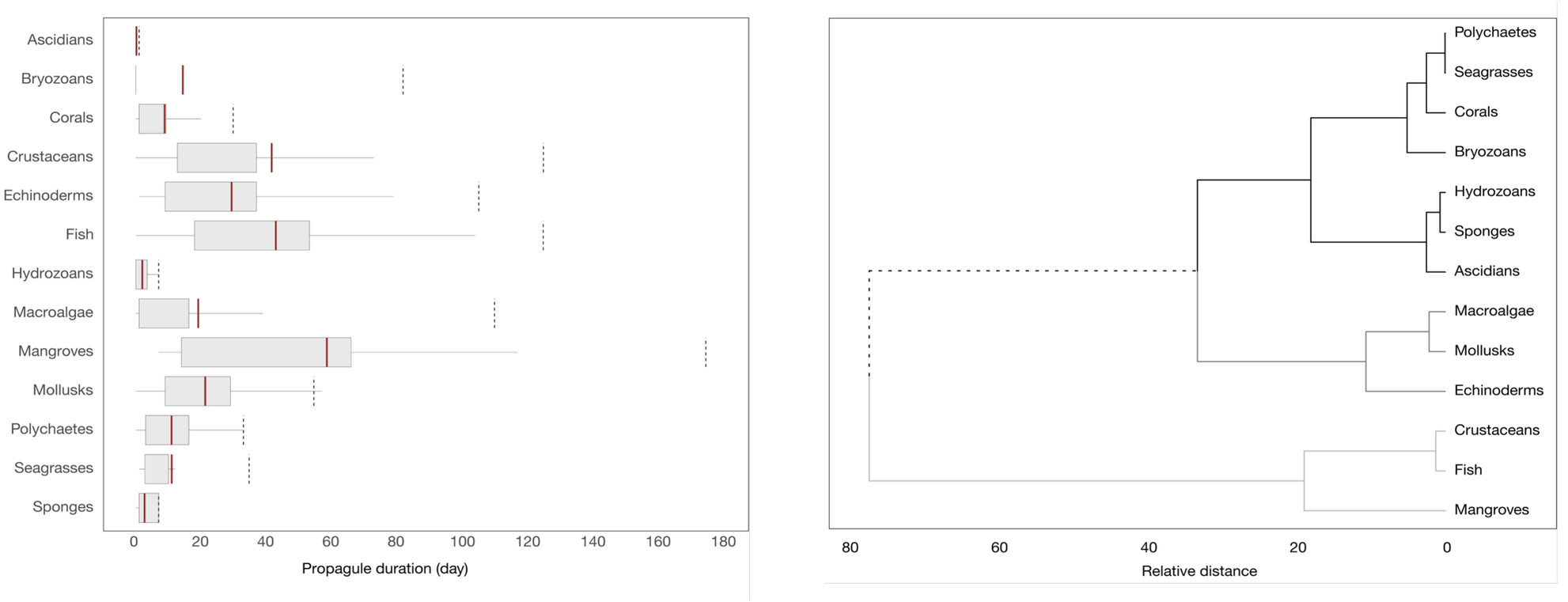
OPEN

DATA DESCRIPTOR

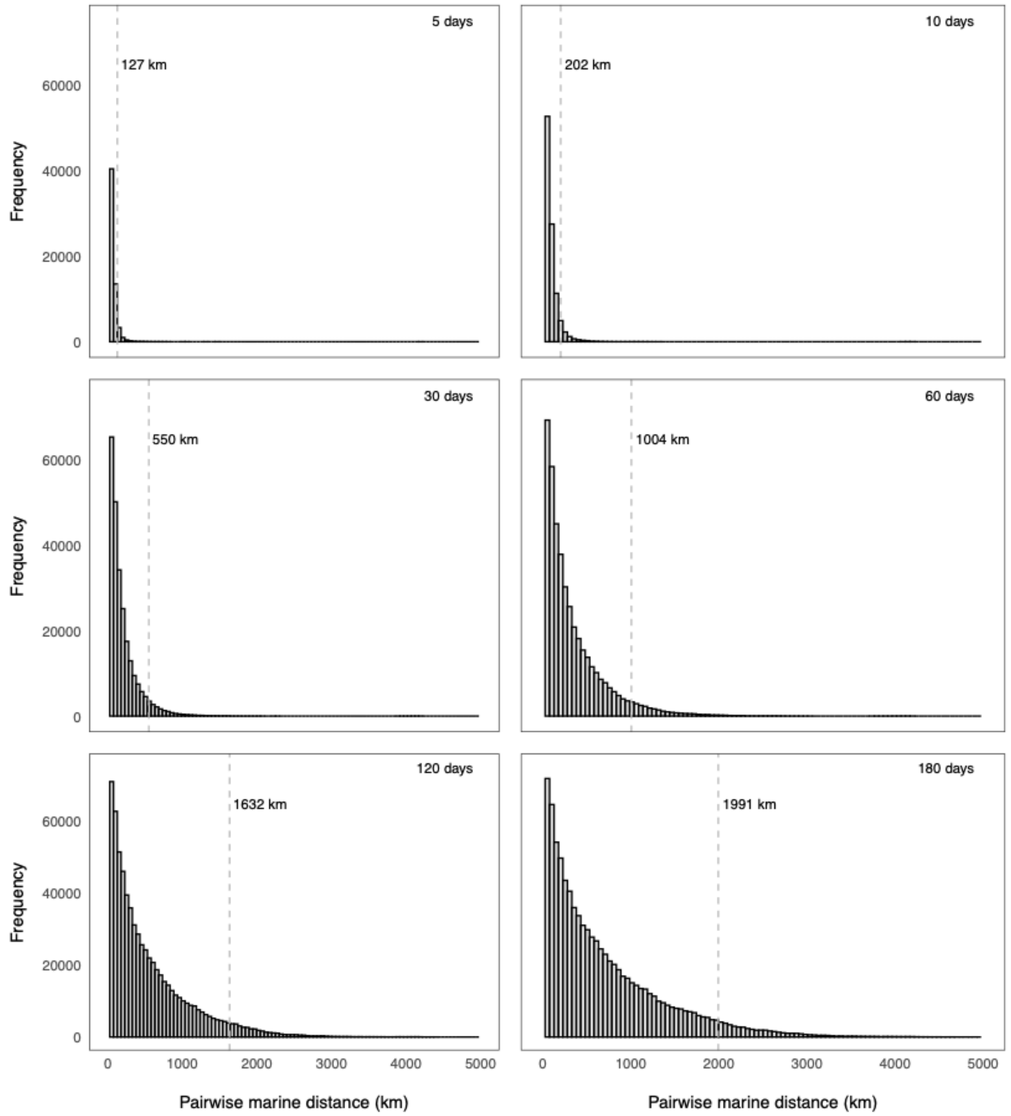
Coastal oceanographic connectivity at the global scale: a dataset of pairwise probabilities and travel times derived from biophysical modeling

Jorge Assis[1,2]✉, Eliza Fragkopoulou[1], Ester A. Serrão[1] & Miguel B. Araújo[3,4,5]

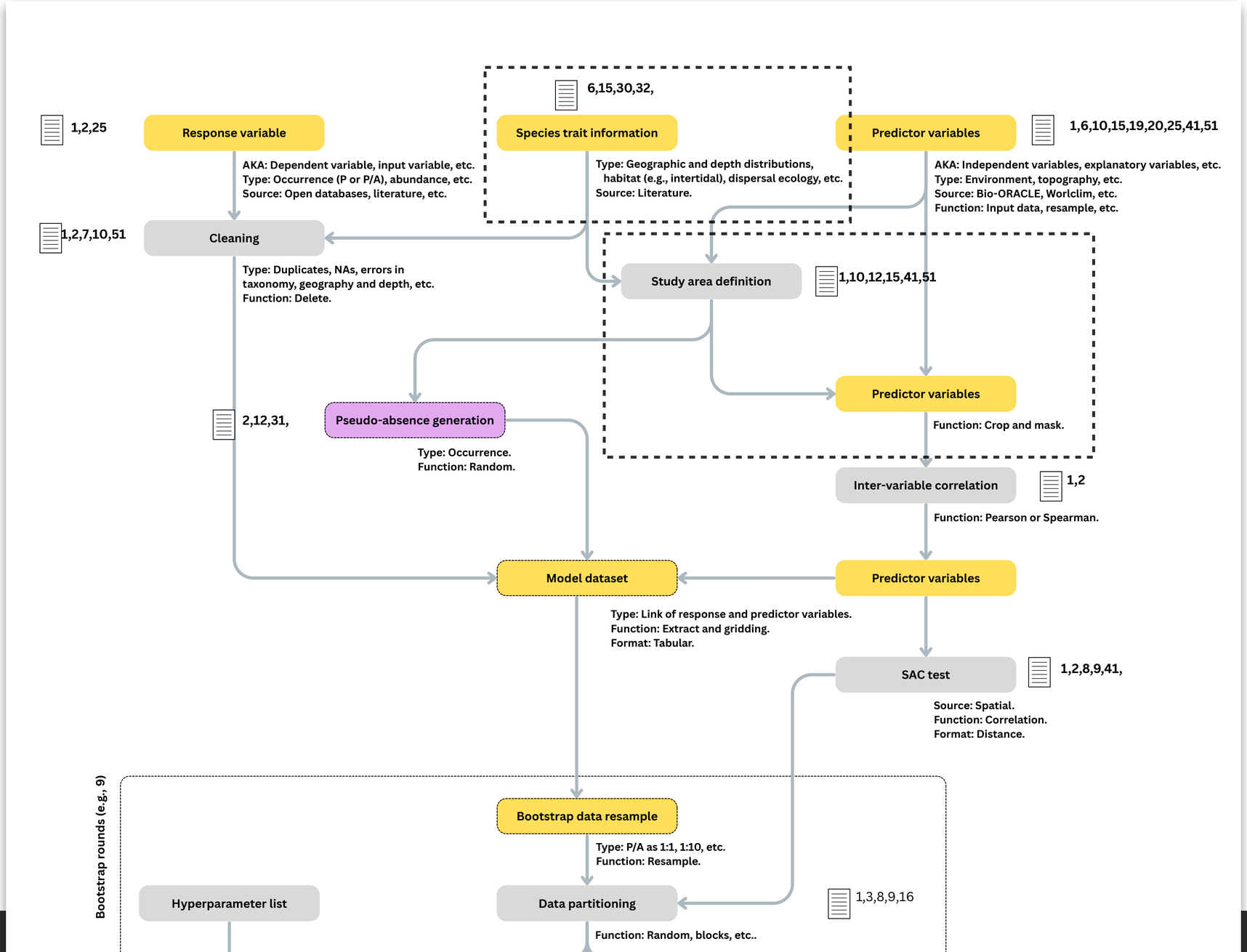
Ocean currents are fundamental drivers of marine biodiversity distribution, mediating the exchange of genetic material and individuals between populations. Their effect ranges from creating barriers that foster isolation to facilitating long-distance dispersal, which is crucial for species expansion and resilience in the face of climate change. Despite the significance of oceanographic connectivity, comprehensive global estimates remain elusive, hindering our understanding of species' dispersal ecology and limiting the development of effective conservation strategies. We present the first dataset of connectivity estimates (including probability of connectivity and travel time) along the world's coastlines. The dataset is derived from Lagrangian simulations of passive dispersal driven by 21 years of ocean current data and can be combined with species' biological traits, including seasonality and duration of planktonic dispersal stages. Alongside, we provide *coastalNet*, an R package designed to streamline access, analysis, and visualization of connectivity estimates. The dataset provides a new benchmark for research in oceanographic connectivity, enabling a deeper exploration of the complex dynamics of coastal marine ecosystems and informing more effective conservation strategies.

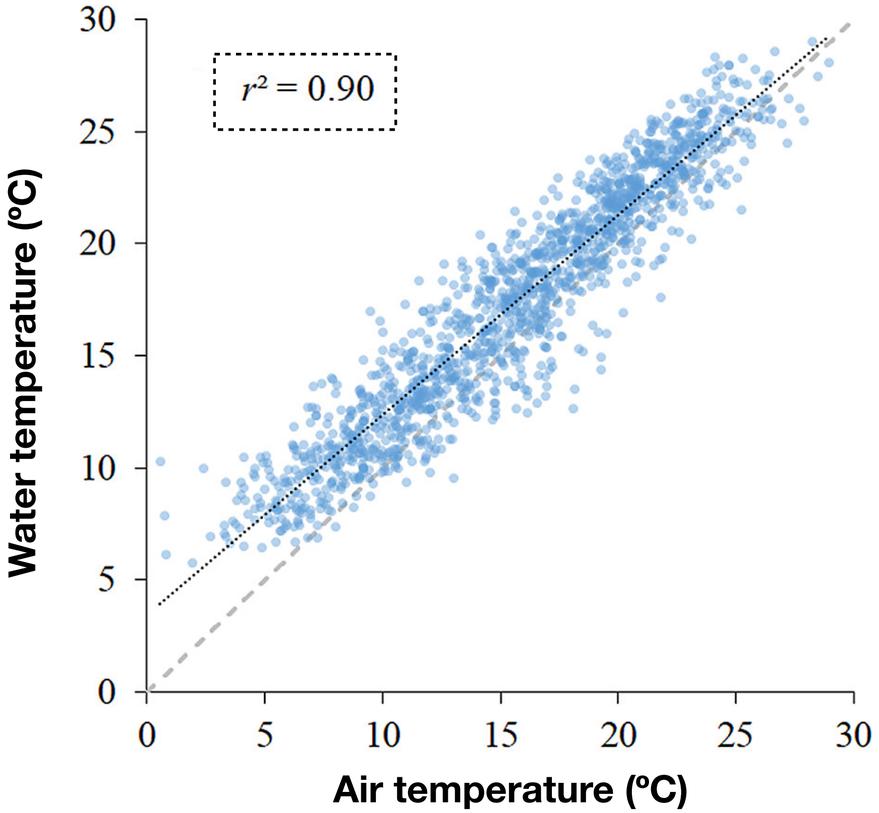


Distribution of **propagule duration periods per ecological group**. Vertical lines depict the mean (red) and the extreme 95th percentile (dashed) of propagule duration across observations. Dendrogram clustering groups based on propagule duration periods.



Frequency of the **distances of connectivity events for different propagule durations**. Vertical dashed line depicts the 99th percentile of connectivity distances.


1,2,25
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Function: Resample.
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Multicollinearity: Many variables are **highly correlated** (e.g., Sea Temperature is often highly correlated with Air Temperature; Depth is correlated with Light Attenuation). **Many algorithms (e.g., GLMs) have problems fitting correlated variables - impeding niche characterization**.

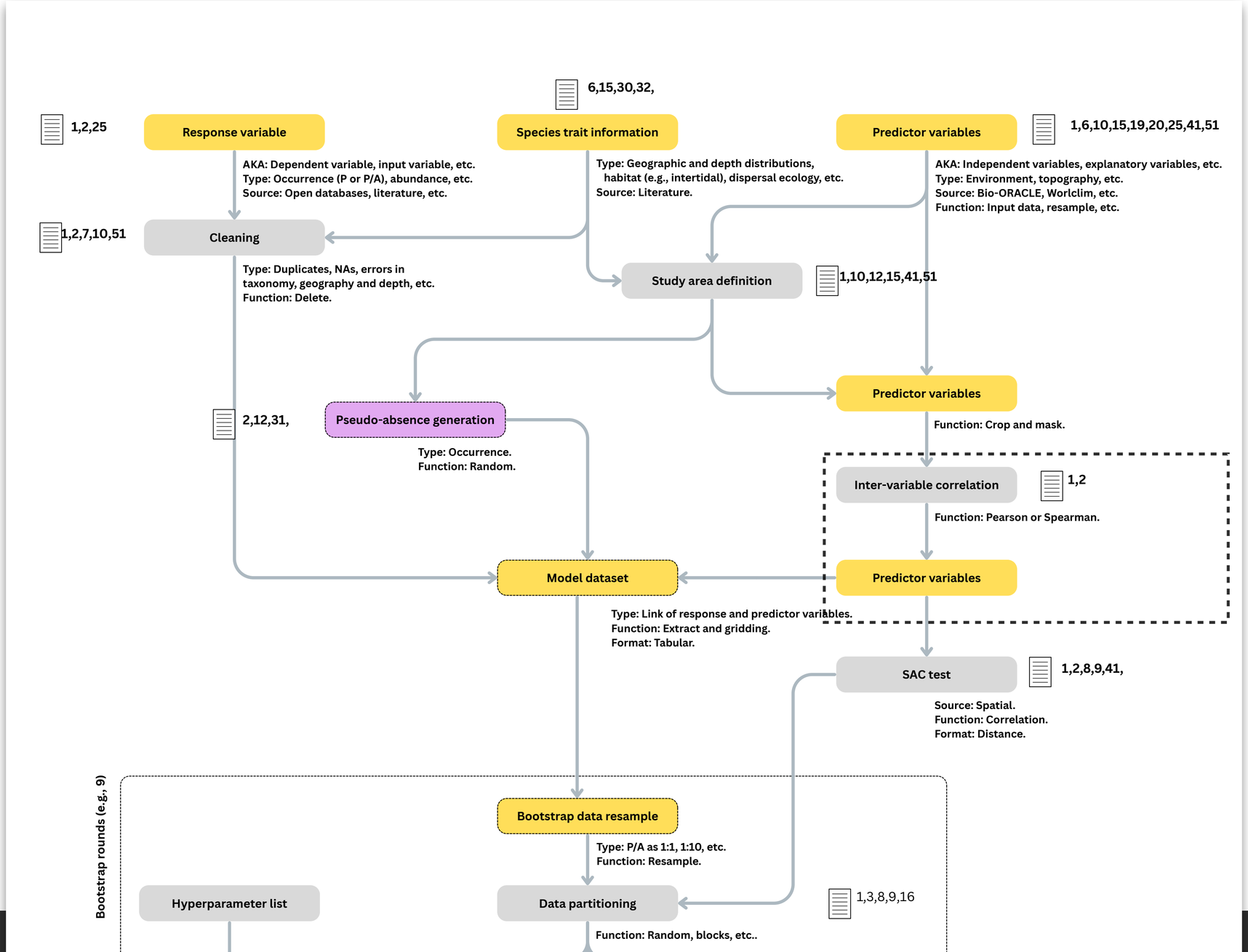
Elith et al. 2009

Correlation Index: The most common metrics used are Pearson's correlation coefficient. Range from -1 (perfect negative correlation) to +1 (perfect positive correlation). In ecological modeling, a common rule of thumb is to flag any pair of variables that $|r| > 0.7$.

Variance Inflation Factor (VIF): VIF can detect and quantify the severity of multicollinearity. A VIF of 1 indicates no correlation. A VIF between 1 and 5 is generally acceptable. A VIF > 10, indicates problematic collinearity.

Ecological Choice: If two variables are highly correlated (e.g., maximum temperature and average temperature), you cannot keep both in the model. The decision of which variable to drop should not be random - **keep the variable that is biologically meaningful on the species' physiology or survival**.

Elith et al. 2009


6,15,30,32,
1,2,25
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Species trait information
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Type: Occurrence (P or P/A), abundance, etc.
Source: Open databases, literature, etc.
Type: Geographic and depth distributions, habitat (e.g., intertidal), dispersal ecology, etc.
Source: Literature.
AKA: Independent variables, explanatory variables, etc.
Type: Environment, topography, etc.
Source: Bio-ORACLE, Worlclim, etc.
Function: Input data, resample, etc.
1,2,7,10,51
Cleaning
Type: Duplicates, NAs, errors in taxonomy, geography and depth, etc.
Function: Delete.
Study area definition
1,10,12,15,41,51
Predictor variables
Function: Crop and mask.
2,12,31,
Pseudo-absence generation
Type: Occurrence.
Function: Random.
Inter-variable correlation
1,2
Function: Pearson or Spearman.
Predictor variables
Model dataset
Type: Link of response and predictor variables.
Function: Extract and gridding.
Format: Tabular.
SAC test
1,2,8,9,41,
Source: Spatial.
Function: Correlation.
Format: Distance.
Bootstrap rounds (e.g., 9)
Bootstrap data resample
Type: P/A as 1:1, 1:10, etc.
Function: Resample.
Hyperparameter list
Data partitioning
1,3,8,9,16
Function: Random, blocks, etc..